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Qualitative assessment of content validity of the modified EQ-HWB-9 by patients, caregivers, and social care users in the United States

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Abstract

Background:

The EQ-HWB-9 is a measure designed to capture health, social care, and caregiver-related quality of life across relevant populations. As part of ongoing development, modifications have been made to item wording, response scales, and layout informed by international collaborations. Qualitative evidence to support the content validity of the modified EQ-HWB-9 in the United States (US) is important to

support its use in this major market. This study evaluated the comprehensibility, relevance, and comprehensiveness of the modified EQ-HWB-9 in US adults.

Methods:

US adults with chronic health conditions, serving as unpaid caregivers, and/or social care users were purposively recruited through an online panel. Participants independently completed an online, pre-interview survey including both experimental and modified EQ-HWB-9 versions and a cognitive interview conducted online using a semi-structured interview guide. Participants compared experimental and modified wording, response scales, layout, provided item-level interpretations and examples, and discussed perceived comprehensiveness of the measure. Interviews were conducted until thematic saturation was achieved. Verbatim transcripts were qualitatively coded and analyzed using the framework method.

Results:

Thirty-four participants (mean age 55; 71% female) completed interviews. Most items demonstrated strong face and content validity; participants readily interpreted daily activities, exhaustion, loneliness, anxiety, sadness/depression, and pain as intended and provided rich, relevant examples. However, the mobility item was identified by several participants as applying only to users of mobility devices. Across modifications, participants preferred “difficulty” over “trouble” (cognition), “lack of control” over “no control,” “a little of the time” over “only occasionally,” and “moderate difficulty” over “some difficulty.” Most participants found repeated question stems clearer than shared item stems. Half of respondents perceived the EQ-HWB-9 as comprehensive, with suggested additions (e.g., sleep, self-care) largely aligning with domains included in the long-form EQ-HWB.

Conclusions:

Findings provide qualitative support for the content validity of the modified EQ-HWB-9 in US populations and reinforce several proposed wording and response-scale refinements. Select issues identified with the wording of the mobility and control items can inform revisions in the wording of a finalized version of the EQ-HWB.

Keywords: EQ-HWB, Content Validity, Caregivers, Chronic Illness,

Declarations***Ethics approval and consent to participate***

This study was approved by the University of Illinois Chicago Institutional Review Board (IRB#2025-0169). Individuals were informed of their ability to refuse participation without consequence. Participants provided electronic informed consent in the online survey, verbal consent prior to the interview, and permission to record the interview. Confidentiality was maintained throughout the research process.

Consent for publication

N/a

Availability of data and materials

The data generated and analyzed during the current study are not publicly available.

Competing interests

MK and ASP are members of the EuroQol Group. JY is supported by the University of Illinois and Pfizer Inc. Fellowship in Health Economics and Outcomes Research 2024-2026. JLN was previously supported by the University of Illinois and Pfizer Inc. Fellowship in Health Economics and Outcomes Research 2021-2023

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Authors' contributions

JLN, ASP, and MK conceptualized the study. JLN conducted data collection and JLN, JY, and MK conducted qualitative analysis. All authors contributed to interpretation of findings. JLN and MK drafted the manuscript, and all authors critically revised it for important intellectual content. All authors approved the final version of the manuscript.

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Background

The EQ Health and Wellbeing (EQ-HWB) instrument is an international measure intended to assess health constructs broadly relevant to patients, social care users,

and caregivers.(1) The EQ-HWB was designed for use across health, social care, and public health settings, enabling estimation of quality-adjusted life-years for evaluating interventions within and across these sectors.(2) Both the long-form EQ-HWB (24 items) and short-form EQ-HWB-9 were developed through literature reviews, item generation, qualitative face validation, and psychometric testing across the US and five other countries (Argentina, Australia, China, Germany, and the United Kingdom). (1, 3, 4)

Since the initial phases of development, the EQ-HWB-9 has demonstrated adequate psychometric performance in the US general public, patients with cancer, and caregivers, including sensitivity to differences in mental health and caregiving types and burden.(4-7) However, establishing qualitative evidence of comprehension, relevance, and comprehensiveness of items is essential, consistent with COSMIN guidance for content validity assessment.(8) Content validity—the degree to which an instrument's content adequately reflects the construct being measured—is considered an essential measurement property of patient-reported outcome measures, as other psychometric properties are only meaningful if the instrument measures the intended construct. This is particularly important in the US context given the increasing role of patient-reported outcomes in regulatory and health technology assessment decisions,(9-11) and the need to ensure instrument content resonates with US respondents across several intended target populations, including patients, caregivers and social care users.

Following initial development of the “experimental version”, modifications to the EQ-HWB-9 have been proposed based on international testing and harmonization efforts to improve comprehension and optimize response scales. These modifications include alternative item wording (e.g., "difficulty" versus "trouble" for

the cognition item; "lack of control" versus "no control"), response scale labels (e.g., "a little of the time" versus "only occasionally" for frequency items; "moderate" versus "some" for difficulty items), and layout changes (a shared item stem versus repeated, complete question text), forming the "modified version". As the EQ-HWB-9 remains in an experimental phase, qualitative evidence is needed to evaluate these proposed modifications prior to finalizing the instrument. Thus far, comprehensive qualitative evidence of content validity of the modified instrument has not been established in US populations. This study is part of a coordinated international effort to evaluate instrument modifications and content validity across multiple countries; the rationale and details of these modifications are described in the introduction to this collection. (Placeholder: Reference Collection Introductory Paper) This study aimed to qualitatively assess the content validity of the modified version of the EQ-HWB-9 in terms of comprehensibility, acceptability, and comprehensiveness through cognitive interviews with US patients, caregivers, and social care users.

Methods

Recruitment

We recruited participants purposively via ResearchMatch, a nationwide volunteer registry for health research established by multiple academic institutions with support from the NIH Clinical Translational Science Award (CTSA) program. This registry maintains a database of volunteers with diverse health and caregiving experiences who have provided consent for researcher contact regarding potential study participation. We screened potential participants using ResearchMatch profile

data and subsequently contacted eligible individuals by email for enrollment confirmation.

Inclusion criteria required participants to be US residents aged 18 or older, English-speaking, and belonging to at least one of the following populations relevant to the EQ-HWB: (1) individuals with chronic conditions - defined as provider-diagnosed physical or mental illness, disability, or impairment persisting for 12+ months; participants selected their conditions from provided disease categories; (2) unpaid caregivers - those providing unpaid, informal assistance with personal care, medical needs, household tasks, finances, or service coordination to an ill, disabled, or elderly adult; weekly caregiving hours were documented; (3) social care recipients - individuals receiving functional support in the previous year through paid/unpaid caregiving, home health services, disability accommodations, residential care, or assistance with personal/medical/transportation needs.

Following guidelines for content validity assessment,(8) we aimed to recruit 7-10 participants per group.(12, 13) Final sample size was guided by thematic saturation,(14) with recruitment continuing until no novel insights emerged from successive interviews.

Pre-Interview Survey

Eligible individuals received a Qualtrics survey, which was completed independently online prior to scheduling the interview. The survey collected demographic and health data to verify eligibility. The survey also included three response option sorting tasks where respondents were presented with the response options from each of the three scales (frequency, severity, difficulty) of the modified EQ-HWB in a random order and were tasked with ordering the options based on their perception

of least to most frequency, severity, or difficulty. All respondents completed three tasks total, one corresponding to each scale. Lastly, respondents completed both the modified and experimental EQ-HWB-9 versions (see supplemental material for each version). Following survey completion, participants scheduled individual interviews in an online calendar system. To minimize fraudulent participation, we distributed unique survey links and cross-referenced responses with ResearchMatch profiles.

Interview Methods

We developed an interview guide based on COSMIN content validity principles⁽⁸⁾ and prior EQ-HWB validation work.^(7, 15) Two pilot interviews were conducted with unaffiliated PhD students and used to inform final refinements to the interview guide.

Semi-structured interviews were conducted online using Zoom, employing a three-phase approach: cognitive interviewing, targeted probing, and open discussion.⁽¹⁶⁾ A single interviewer (JLN), a doctoral candidate with qualitative research training and clinical pharmacy experience treating diverse chronic conditions, conducted all sessions. Apart from scheduling correspondence, no prior relationships existed with participants. Interviews were conducted in English.

Interviews began with a study overview, procedural explanation, and verbal consent including recording permission. The modified EQ-HWB-9 content validity discussion comprised the primary focus, addressing item comprehension, relevance, and modification preferences. Interviews lasted approximately 60-90 minutes. Upon completion, participants received \$50 digital gift cards. Field notes documenting observations and impressions were recorded immediately post-interview.

EQ-HWB-9 Content Validation and Modifications Testing

The interview protocol examined the content validity of each of the nine items in the EQ-HWB-9. Items were assessed individually in the order they appear in the measure. For each item, respondents were asked to explain their interpretation of the presented item and provide examples from their own life, including relevant examples for specified response options.

Additionally, the interview examined several modifications to the EQ-HWB-9 based on prior psychometric testing and international harmonization efforts. Participants completed both the experimental version and the modified version containing the following changes:

Item Wording Modifications: We tested alternative phrasing for four items: (1) Cognition - "**difficulty** concentrating/thinking clearly" versus "**trouble** concentrating/thinking clearly"; (2) Control - "**lack** of control over your day-to-day life " versus "**no** control over your day-to-day life "; (3) Mobility - "getting around inside **and** outside" versus "getting around inside **or** outside"; and (4) Item ordering - presenting mobility before daily activities versus the reverse sequence.

Response Scale Modifications: Two response option changes were evaluated: (1) For the frequency scale (applied to exhaustion, loneliness, anxiety, and sad/depressed items) - "a little of the time" versus "only occasionally"; and (2) For the difficulty scale (applied to daily activities and mobility items) - "moderate difficulty" versus "some difficulty."

Layout Modification: We compared two presentation formats with the daily activities and mobility items: a single shared stem ("In the past 7 days, how much difficulty did you have...") with items listed below, versus complete questions

repeated for each item (e.g., "In the past 7 days, how much difficulty did you have doing day-to-day activities?").

During interviews, participants were shown both versions side-by-side and asked to: identify differences between versions, explain their interpretation of each modification, state preferences with rationale, and discuss whether modifications affected their response selection. The interview guide included standardized prompts to ensure consistent exploration of each modification across all participants.

Analytic Approach

Interview recordings were transcribed verbatim and de-identified by a member of the research team, beginning with Zoom auto-transcription and verifying against recorded audio. Secure cloud-based storage housed audio files accessible only to research personnel. We employed framework analysis combining deductive and inductive approaches to develop and refine coding structures for cross-case comparison.(17, 18)

Initial familiarization involved reviewing transcripts alongside field notes. Line-by-line coding identified salient features. The preliminary framework addressed: item comprehension, relevance perceptions, modification preferences, and measure completeness, with additional codes emerging inductively. Three initial transcripts underwent dual coding (JY, MK) with team discussion establishing consensus. Subsequent transcripts were coded individually (JY, MK, or JLN) with secondary review (MK or JLN). Weekly team meetings facilitated refined definitions and frameworks. Analysis was conducted in Microsoft Excel (Microsoft Corporation). Representative quotations were extracted and organized in the analytic matrix.

Saturation was monitored weekly to confirm no emergent themes appeared in final interviews, providing justification that no further interviews were required. (14) The finalized framework was applied after all coding. Participants did not review transcripts or findings. We followed the COREQ checklist to report essential elements of qualitative research (supplemental material).(19)

Results

Survey Results

Participant Characteristics

In total, 757 individuals were contacted through ResearchMatch and 86 responded and received the online survey via email. Of these, 40 completed the survey, 3 explicitly declined participation, 5 were no-shows for scheduled interviews, and 1 was excluded due to inconsistencies between their survey responses and ResearchMatch profile. Thus, 34 participants completed the qualitative interview and were included in the analytic sample, achieving thematic saturation.

In terms of age, participants ranged from 25 to 81 years old (mean = 55; SD = 18;

Table 1). The sample was predominantly (71%) female. Participants were allowed to select more than one category for race and ethnicity and 27 identified as White, 3 as Hispanic, 2 as Black or African American, 1 as Asian, and 2 as Mixed Race. In terms of education, 25 participants (74%) reported having at least a bachelor's degree. Participants were relatively evenly dispersed across US census regions. While all but two participants reported chronic health conditions, the final categorization of participants occurred after confirming their details in the interview, so participants endorsing being caregivers (32%) or social care users (24%) were classified as such to guide interviews and analysis.

Table 1: Participant Characteristics***Response Option Sorting Task***

In the response option sorting task, $n=27$ (79%), $n=23$ (68%), and $n=29$ (85%) correctly ordered the difficulty, frequency, and severity labels, respectively (Supplementary Material). Some participants fully reversed the order of the ranking ($n=3$ for difficulty, $n=2$ for frequency, $n=2$ for severity); this was interpreted as a misunderstanding of the task instructions rather than the response options and was counted as correctly ordered.

EQ-HWB-9 item Face and Content Validity

Interpretations of item meaning and relevance are summarized in **Table 2**.

Participant quotes are identified with an assigned number, age, gender (M, male; F, female; NB/O, non-binary/other), and classification (chronic illness, caregiver, social care user).

Table 2: EQ-HWB-9 (modified version) Assessment of Content Validity

Daily Activities: All participants ($n=34$) understood the intended meaning of “doing daily activities” as a broad set of routine tasks spanning the specified examples (working, shopping, housework) and provided additional examples of taking care of pets, gardening, exercise, activities of daily living like bathing and dressing, and role-specific responsibilities (managing caregiver or care-recipient schedules), which most ($n=30$) explicitly endorsed as relevant to their lives:

“Like it says, kind of your day to day... am I gonna go to the gym? Can I go grocery shopping, run errands, go to the ATM?” (#025, 37M, Chronic Illness)

Interpretations of difficulty emphasized prioritization under limited time and energy, balancing activities that “have to get done” versus optional activities that “can

wait". Participants also considered the physical and mental effort required for their daily activities and competing priorities. Caregivers and social care users also provided examples contextualized to their situations, such as planning around caregiving schedules:

"I also had to plan around my mother's schedule... if I had to go pick up prescriptions for her, was I gonna bring her with me? Did she need other things besides what I had planned?" (#024, 48F, Caregiver)

Mobility: Most participants (n=25) understood the intended concept being assessed. However, compared to other items, more participants (n=9) had difficulty fully understanding the extent of the experiences the item was intending to capture. Consequently, almost half of all participants (n=14) were not certain whether the item was relevant to them. The parenthetical statement included after the item "(using e.g., a walking stick or wheelchair if you normally use them)" led several participants (n=5) to incorrectly assume that the item only applied to users of mobility devices:

"Luckily I don't have to utilize, you know, some type of device to get around... I don't know what else you want me to say about it" (#030, 47F, Chronic Illness)

The term "walking stick" was also not as familiar as other mobility devices such as canes or walkers. Participants also noted ambiguity about the extent to which they should consider their mobility devices when responding:

"I have a power wheelchair [electric wheelchair] that I drive independently, so certainly if that was unable to be used, that would create a lot of difficulty for me. I do have a lot of trouble pushing my manual wheelchair any kind of long

distance, so that would really restrict my movement quite a bit." (#012, 46F, Social Care User)

Similarly, participants also questioned whether the item included driving or other modes of transportation and noted that they faced different obstacles inside (stairs, carpet) versus outside (uneven terrain, inadequate accessibility of wheelchair ramps or elevators). For caregivers, it was often not immediately clear if the item was relevant because they did not consider the mobility difficulties of their care-recipients to affect their own mobility.

Exhausted: Nearly all participants (n=33) understood the item as intended and most (n=29) found it relevant. Participants described experiences of depleted physical or mental energy, whether from balancing caregiving with their own daily needs, illness, physical labor (including exercise), or emotional strain. Those unsure of the relevance of the item noted that "exhausted" felt too extreme or situational (e.g., only felt after vigorous exercise or with an acute illness) and preferred a term such as "fatigue".

"[The] meaning is not necessarily clear...I could just come back from a very intense workout and be exhausted because of that. I wonder if there would be a different way of putting that?" (#001, 81F, Chronic Illness)

Lonely: Almost every participant (n=33) understood what the item was asking, even if they found it difficult to rate. The experience of "feeling lonely" was understood to be alleviated by not just the number of social contacts, but also the quality of those interactions. Caregivers and social care users receiving caregiving noted feeling lonely and "unseen" even though they spent the day with their care-recipient or caregiver.

"[I] still feel lonely despite having people around for me: it depends on the quality of the relationship with the caregiver... while I'm eating lunch, they're not talking to me or anything. They're just sitting there...playing on their phone... And so, even though there's somebody with me, I still feel very alone in those moments." (#012, 46F, Social Care User)

Most (n=27) viewed loneliness as an important dimension of mental health and caregiving burden:

"Very important question...especially for seniors: we tend to be isolated"
(#009, 79F, Social Care User)

However, approximately 25% of participants (n=7) noted that introverted personalities, enjoying solitude, or having strong social support made the item feel less relevant to them.

Concentrating or Thinking Clearly: Most participants (n=31) comprehended this item, though difficulties arose from distinguishing between difficulty concentrating (focus, attention) versus thinking clearly ("brain fog") and selecting one response for both experiences:

"Thinking clearly is kind of vague, but concentrating is pretty easy for me to interpret... [they] seem different." (#017, 37NB/O, Social Care User)

Most (n=29) found it relevant, explicitly linking cognitive difficulties to aging, exhaustion, pain, or stress rather than viewing them as independent issues:

"I used to be kind of sharp... it's either some combination of the impact from the chemo, possibly, or just I'm getting old. I just don't have the memory that I used to; I can't think fast anymore." (#028, 68M, Chronic Illness)

Anxious:

Nearly all participants (n=33) understood mental and/or physical manifestations of anxiousness, including worry, racing thoughts, restlessness, or heart palpitations, often tied to specific triggers (traffic jams, medical uncertainties) or more generalized anticipation of future events.

"...At doctor's appointments, I will feel super anxious for results on testing for her, not knowing if we're gonna have positive results, negative results; it kind of weighs a lot in your chest." (#006, 38F, Caregiver)

Participants (n=24) generally endorsed "feeling anxious" as relevant to capture. Those who found the item less relevant noted it did not personally apply, often from conflating the feeling of anxiousness with a clinical diagnosis of an anxiety disorder.

Sad/Depressed: Similar to the item assessing feeling anxious, most participants (n=31) understood the intended meaning of the item, with some focusing on a clinical diagnosis of depression. There was ambiguity related to including both the terms "sad" and "depressed", as sadness was described as transient and tied to specific events while depression was described as a deeper, longer-lasting state:

"That is a bad question because sad and depressed are two different things: I can feel sad because a friend of mine just died. That doesn't mean I'm depressed. I'm just sad. or I can be depressed without having a reason for sadness. Those two actually do not belong together."(#001, 81F, Chronic Illness)

Control: Compared to other items, fewer participants (n=27) clearly understood the intended meaning of this item. Participants described diverse experiences with

feeling helpless or hopeless related to injury or illness (and subsequent treatment) for themselves or a loved-one:

"I [consider] insurance costs, medicine availability, whether there's actually anything that treats what you're feeling... my medicine stopped working because my body sort of rejects them... you also have to worry about if your doctor is going to listen to you or understand what you're saying." (#014, 72F, Chronic Illness)

Particularly, participants with chronic illness rather than caregiving or social care experience tended to relate a loss of control to more extreme situations, ranging from situations of domestic abuse, being incapacitated, or only applicable to those receiving social care services:

"If I were in a nursing home, or [an] adult care facility, prison, or somewhere where I had essentially lost my freedom, this would apply." (#010, 76M, Chronic Illness)

While the item resonated less strongly for participants with chronic illness, caregivers and social care users particularly resonated with feelings of loss of autonomy in their daily life:

"He [husband, care-recipient] had to have surgery 150 miles away. So I'm living in a hotel for Christmas Day, eating at Wawa [convenience store]. I was away from my house, away from my cat. I had no control over what was happening to him in the hospital. I had no control over my life." (#007, 70F, Caregiver)

Pain: Nearly all participants (n=33) understood this item and could relate to painful experiences from illness or injury. Caregivers without physical strain from their caregiving duties found this item less applicable to their situation:

"I think it could certainly be relevant to some people. For me, and the caregiving I provide, I don't have to do a lot of physical stuff being that she's mobile on her own." (#032, 46M, Caregiver)

Sources of ambiguity included the subjectivity of pain severity, responding when pain fluctuated during the week, and relating to pain in terms of severity instead of frequency.

EQ-HWB-9 Modifications Face Validity

The face validity of the instrument modifications are summarized in Table 3.

Table 3: Modified Items Assessment

Item Wording Modifications

For the cognition item, participants showed a clear preference for "difficulty" (n=15) over "trouble" (n=8), with n=13 viewing them as meaningfully different. Those preferring "difficulty" characterized it as more clinical and appropriate for health assessment, while "trouble" was seen as too casual or vague:

"Difficulty might lean more toward a medical reason. Trouble, maybe not so much" (#003, 80F, Chronic Illness)

With the control item, participants indicated strong preferences for "lack of control" (n=20) versus "no control" (n=6), with 28 participants distinguishing between the terms. Participants viewed "no control" as absolute and extreme, while "lack" allowed for degrees of autonomy. This distinction was particularly important given

the response scale options, with one participant noting that "no control" seemed more suited to a binary, yes/no responses.

"I Love the word 'lack'...If I lacked control, I can regain it some way. If you ask me if I had no control... I would give you a more blunt answer. And you probably wouldn't even have to give me options: it would be a yes or no.
(#031, 43M, Caregiver)

Mobility wording ("inside **and** outside" versus "inside **or** outside") generated mixed responses, with n=8 preferring "and," n=8 preferring "or," and n=19 interpreting them as distinct. Participants recognized that indoor and outdoor environments pose distinct challenges, leading some to suggest separate questions or "and/or" wording.

Response Scale Modifications

"A little of the time" was preferred (n=15) over "only occasionally" (n=9) for frequency items, with n=16 interpreting them differently. Participants found "a little of the time" more quantifiable within the 7-day recall period, while "occasionally" seemed to imply longer timeframes or was too similar to "sometimes."

For difficulty ratings, "moderate" (n=17) was favored over "some" (n=7), with n=19 interpreting them as distinct. "Moderate" provided clearer severity indication and better aligned with clinical assessments. Participants noted that "some" was too vague and potentially overlapped with "slight" difficulty.

Layout Modification

Removing the shared item stem was preferred by n=11 participants, while n=5 preferred a shared item stem, and n=7 had no preference. Those favoring individual

complete questions found them clearer and easier to process without needing to remember the stem.

Item Order

There was no consensus on whether mobility or daily activities should appear first, with arguments for logical flow in either case. Those favoring mobility first endorsed that day-to-day activities depend on mobility; others used the same logic to support asking about day-to-day activities first. Ultimately, no participant had very strong opinions about item ordering.

EQ-HWB-9 Comprehensiveness: Discussion of Missing Content

Results pertaining to comprehensiveness are summarized in the Supplementary Material.

Half of participants felt the EQ-HWB-9 was comprehensive as presented. The most frequently suggested additions to address gaps in the measure were: additional mental/emotional health items, including by separating mental versus physical exhaustion and distinguishing sadness from depression (n=4); overall health assessment scales (n=3) to synthesize the nine items; and environmental factors (n=3), such as financial circumstances and resource access.

Discussion

This study provided evidence to support the content validity of revisions to the modified EQ-HWB-9 in the US, while also comparing proposed modifications with the experimental version of the measure. Participant experiences as patients, caregiver, and social care users provided evidence that the items comprising the modified EQ-HWB-9 resonate with US participants belonging to some of the relevant target

populations of the instrument. The majority of items elicited relevant, well-articulated examples from participants' own lives. Most interpretations were aligned with the nuances the instrument intends to capture, including daily activities encompassing daily responsibilities, self-care, and role-specific (e.g., caregiving) tasks, exhaustion as both physical and mental depletion, and loneliness as relating to quality of social connections and their impact rather than simply frequency of contact.

Items elicited different examples and interpretations of relevance between groups, reflecting sensitivity to distinct lived experiences. Caregivers and social care users particularly related more with the item assessing lack of control than did participants with chronic illness alone. Conversely, caregivers without concomitant physical illness themselves found items assessing mobility and pain to be less relevant, highlighting the importance of emphasizing the respondent perspective in the instrument instructions, particularly when the EQ-HWB-9 is used in caregiver populations. Comparing the modified and experimental versions of the EQ-HWB-9, participants expressed mixed opinions that either favored the modification or were ambivalent without strong preferences, even when noting differences between experimental and modified wording in terms of meaning, nuance, or accessibility. Rich, qualitative evidence enabled subtle insights into perceived differences between modified and experimental wording in terms of connotation and approachability for respondents of different regional and educational backgrounds, for example, the accessibility of words like "trouble" and "some" to individuals with less formal education. These results complement quantitative findings from the response option sorting tasks for the modified difficulty, frequency, and severity scale labels, which were incorrectly sorted by approximately a quarter of

participants. Collectively, this evidence supports synthesizing elements of the modified and experimental versions as the foundation for further refinements to finalize the EQ-HWB-9.

Still, some common difficulties occurred with select items and proposed modifications, warranting further evaluation and potential refinement. While most modifications were preferred or equivocal compared to the experimental version, the use of a shared item stem for the mobility and activities items was an exception, with more participants finding repetition of question text for each item to be clearer. The mobility item especially presented comprehension challenges, with nearly half of participants expressing uncertainty whether it was relevant to them. Much of this confusion stemmed from the parenthetical example "(using e.g., a walking stick or wheelchair if you normally use them)," which led several participants to assume the item applied only to users of mobility devices, a finding similarly reported in another content validity study of the EQ-HWB.(20) Similarly, the control item generated confusion related to its parenthetical example "(e.g., no choice in what you do or when you do it)," which some participants perceived as contradicting the modified wording of "lack of control" used in the item. Survey literature recognizes that examples can be useful in certain circumstances to clarify items but can also cause focusing effects resulting in narrower interpretations.(21) The necessity of parenthetical examples should be revisited while finalizing the instrument, particularly if other country investigators note similar issues. Other aspects to address include cultural and linguistic adaptation from UK English to US English, for example, changing "walking stick" to a more familiar term for US respondents, such as "cane".(22-24)

Regarding comprehensiveness, approximately half of participants felt the EQ-HWB-9 adequately covered relevant aspects of health and wellbeing, while others identified room for additional potential domains such as sleep quality, mental/emotional specificity, and social support. Notably, these suggested additions are largely captured in the long-form EQ-HWB, supporting the purpose of the short-form as a brief, preference-weighted measure. Interviews revealed deep, genuine engagement from participants who recognized the value and need for a measure capturing health and wellbeing across diverse circumstances. This aligns with a growing trend toward measuring broader aspects of health and wellbeing to incorporate into value discussions and decision-making beyond traditional dimensions of health-related quality of life.(25, 26)

[In discussing the comprehensiveness of the instrument](#), multiple participants also suggested adding a global assessment item to synthesize the nine items. These suggestions may have been influenced by earlier discussion of potential visual analogue scale (VAS) versions for inclusion alongside the EQ-HWB-9, similar to the approach used with existing EQ-5D instruments (reports reported elsewhere).(27) However, participants were not explicitly prompted to connect consideration of a VAS with their assessment of the comprehensiveness of the EQ-HWB-9, warranting further exploration of whether a VAS alongside the EQ-HWB-9 satisfies preferences for a global assessment item. Informed by the incorrect ordering observed in the response options sorting task, further quantitative work is planned to determine whether these preliminary findings are replicable in a larger sample to inform finalizing the choice of response option labels for the three scales presented in the measure.

There are some limitations to qualitative evidence, as it should be interpreted without the assumption that findings are representative of the range of interpretations of the EQ-HWB-9 by other respondents in the US. Compared to the makeup of the US population, this sample included more woman than men, fewer racial or ethnic minorities, and most individuals held a college or advanced degree, reflecting the usual composition of online research panels. These characteristics may be linked to potential differences in item interpretations. It is uncertain how well the items would be understood by individuals with less formal education, lower literacy, or who speak English as a second language. Further, specific patient groups were not purposively sampled, preventing insights into unique or diverging interpretations across disease states. Still, the sample included participants across a wide range of ages and with diagnoses including inflammatory conditions, depression, and cancer. There was significant overlap in classifying participants as patients, caregivers, or social care users, with most participants identifying with more than one group, for example, providing caregiving while experiencing a chronic illness. Therefore, findings specific to one respondent group could not readily be disentangled. As the instrument becomes utilized in real world contexts in diverse US populations, mixed method and quantitative psychometric evidence, including differential item functioning, should be assessed across intended target groups and in lower educated and culturally diverse populations.

Conclusions

In summary, this study provided evidence of qualitative content validity for the EQ-HWB-9 in US populations, complementing existing quantitative psychometric evidence. The instrument is comprehended and relevant across intended target populations, with findings supporting adoption of most proposed modifications.

Challenges identified with the use of a shared item stem and mobility and control items, particularly the use of parenthetical examples narrowing interpretations, warrant further consideration to finalize the instrument.

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Table 1: Participant Characteristics

	Mean/Count	SD/Percent
Average Age	55.4	18.1
Age Range	25 - 81	-
Age Categories		

	18-34	5	14.7
	35-64	14	41.2
	≥65	15	44.1
Gender			
	Male	8	23.5
	Female	24	70.6
	Non-binary or Other	2	5.9
Race^a			
	White	27	77.1
	Black	2	5.7
	Asian	1	2.9
	Mixed Race	2	5.7
	Hispanic	3	8.6
Highest Level of Education			
	Graduate or Professional Degree	16	47.1
	Bachelor's degree	9	26.5
	Associate degree	6	17.6
	Non-college trade or technical school	1	2.9
	Some college but no degree	2	5.9
US Census Region			
	Northeast	6	17.6
	Midwest	11	32.4
	South	9	26.5
	West	8	23.5
Inclusion Category^b			
	Chronic Illness	15	44.1
	Caregiver	11	32.4
	Social Care User	8	23.5
Number of Reported Chronic Conditions			
	1-3	16	47.1
	4-7	12	35.3
	8-11	6	17.6
General Health Rating			
	Very Good	2	5.9
	Good	13	38.2
	Fair	13	38.2
	Bad	6	17.6
	Very Bad	0	0.0
Caregiving Hours per Week			

1-19	8	23.5
20-49	3	8.8
≥50	1	2.9
No Response	1	2.9
Primary Daily Activity^a		
Looking for Work	1	2.9
Employed	16	47.1
Retired	13	38.2
On Sick Leave or Disability	5	14.7
Student	4	11.8
Caring for Home or Family	2	5.9

a. Participants could select more than one category

b. While participants could select more than one category, they were assigned to one primary category determining the focus of the interview

Table 2: EQ-HWB-9 (modified version) Assessment of Content Validity

EQ HWB-9 Item	Understanding of modified item (Comprehension)	Relevance	Themes and Examples Related to Comprehension and Relevance	Sources of Ambiguity
Daily Activities	Yes (n=34) Difficulty Understanding (n=0)	Yes (n=30) Partial or Uncertain (n=4)	Daily physical functioning, encompassing the provided examples, as well as activities of daily living, taking care of pets/walking dogs, exercising, gardening, and other tasks or errands to care for themselves, others, or their living environment For social care users and caregivers, could also include managing schedules for caregivers/care recipients and/or transportation	The nature of work being discussed, for example, if it involves moving and physical activity versus sitting at a computer (#033) Ability to complete daily activities fluctuate with health condition, making it difficult to select one response (#033)
Mobility	Yes (n=25) Difficulty Understanding (n=9)	Yes (n=20) Partial or Uncertain (n=14)	Key elements of mobility including walking, bending, getting in and out of a car, lifting (e.g., helping care recipients) that allow one to move around their environment	Examples provided within parent interview "using e.g., walking stick, wheelchair if normally used seem to indicate item is relevant"

			Considers obstacles encountered inside versus outside, such as fall/tripping hazards (rugs, chords, stairs), uneven terrain (grass, gravel) and disability access (wheelchair ramps, elevators)	to those who use mobility devices (#001, #020, #030, #031) Walking sticks as recognition of other mobility devices, such as cane or walker (#033) Unclear limit on transportation options or mobility devices the item refers to, for example driving a car, riding a motorcycle, using a motorized wheelchair For caregivers, ambiguity if/when this item relates to their own mobility being affected versus considering the mobility of care recipients having to help recipients navigate their living environment and out of a home, etc. affects the caregiver's freedom of movement
Exhausted	Yes (n=33) Difficulty Understanding (n=1)	Yes (n=29) Partial or Uncertain (n=5)	Encompassing both physical depletion (e.g., muscle weakness, reduced capacity for daily tasks, fatigue) and mental/emotional drain (e.g., stress, lack of motivation,	One person suggested 'exhausted' was only something after intense exertion like vigorous exercise

			burden of caregiving, “mental exhaustion”) Caregivers described exhaustion from balancing their personal needs with those of their care recipient	and not the ongoing, low state as inter recommended fatigue as a alternative to describe the intended con
Lonely	Yes (n=33) Difficulty Understanding (n=1)	Yes (n=27) Partial or Uncertain (n=7)	Participants repeatedly stressed the possibility of being “alone” without being lonely, and vice versa. Mitigated by not just the number of social contacts, but also the quality of interactions. Caregivers could feel lonely and “unseen” even while spending time with their care-recipient.	One participant interpreted t as asking ab physically be alone rather the intended of emotional disconnect
Concentrating or Thinking Clearly	Yes (n=31) Difficulty Understanding (n=3)	Yes (n=29) No (n=1) Partial or Uncertain (n=4)	Difficulty concentrating was related to focus, attention, ability to multitask, or being easily distracted, while difficulty thinking clearly was most commonly described as “brain fog” (#011, #014, #016, #024, #025, #026, #033). Older participants felt this item was relevant to issues with memory and cognition associated with aging. This item was often explicitly attributed to other issues like sleep or exhaustion (#016, #025, #029, #032), pain (#002), or stress and anxiety (#013, #024, #026).	Due to differ in perception meaning of concentrating versus thinki clearly, some this question difficult to ar (e.g., #005, #017)
Anxious	Yes (n=33) Difficulty Understanding (n=1)	Yes (n=24) Partial or Uncertain (n=10)	Considered both physical and mental manifestations of anxiousness including worry, racing thoughts, restlessness,	There was so uncertainty i item was refe to a clinical diagnosis of

			heart palpitations, rapid breathing. Anxiety was tied to specific triggers (traffic jams, medical uncertainties) as well as generalized worry (“reading the news,” “future events”).	“anxiety” very much a perception of anxiousness
Sad/ Depressed	Yes (n=31) Difficulty Understanding (n=3)	Yes (n=23) Partial or Uncertain (n=11)	Sadness described as transient and tied to specific events, ranging from mundane (“store missing your favorite ice cream flavor”, #009) to the death of a loved one. Depression was described as a deeper, longer-lasting state, possibly requiring clinical intervention.	Uncertainty and inconsistency around whether report on even “sad” moments their more serious “depressed” experiences, somehow avoid their responses
Control	Yes (n=27) Partial or Uncertain (n=7)	Yes (n=21) Partial or Uncertain (n=13)	Wide ranging description of experiences from feelings of helplessness or hopelessness related to physical or mental illness or treatment (e.g., chemotherapy for cancer), the inability to play sports because of a past injury, situations of domestic abuse or manipulation by a spouse. For caregivers, also included the burden of being a caregiver and a sense of duty or obligation to put the needs of a care recipient above their own needs, especially if a spouse or family member	Could be considered less relevant for people with chronic illness; commonly viewed from perspective of living in a residential care facility or receiving other social care services (#010, #029); others viewed it in relation to physical limitations (#011) or mental illness including anxiety/depression (#013).

Pain	Yes (n=33) Partial or Uncertain (n=1)	Yes (n=30) Partial or Uncertain (n=4)	Participants described varied pain experiences related to their illnesses or from the burden and physicality of caregiving; n=6 participants specifically mentioned the inherent subjectivity in assessing pain severity (#001, #007, #008, #014, #016, #017)	Some (n=4) caregivers did not consider the item relevant because they did not experience pain associated with their caregiving experience Several participants mentioned some ambiguity about answering for that fluctuates throughout the week, and responded considering both frequency and severity; n=3 participant did not fully understand the item as severity only preference thinking of pain frequency
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Table 3. Modified items Assessment

Modification	Assessment of the Modification	Preference	Participant Feedback	Additional Examples
Getting around inside 'or' (M) (vs 'and' (E)) outside	'And' = 'Or' (n=7) 'And' ≠ 'Or' (n=19) Unclear or Missing (n=8)	And (n=8) Or (n=8) No Preference (n=11) Missing (n=7)	Participants noted differences in 'and' vs 'or' because they could encounter different difficulties in each environment (e.g., fall/tripping hazards (rugs, chords, stairs), uneven terrain (grass, gravel) and disability access (wheelchair ramps, elevators); N=7 participants	"You know I probably had a question about it. Like, inside, but not outside? I would overthink [and]" (Caregiver)

			suggested that these could be separate questions or use wording of 'and/or' on account of these differences. Although differences were noted, preferences and reasons for those preferences were not consistent or able to be articulated clearly	
'Difficulty concentrating/thinking clearly" (M) (vs 'Trouble concentrating/thinking clearly" (E))	Difficulty = Trouble (n=21) Difficulty ≠ Trouble (n=13)	Difficulty (n=15) Trouble (n=8) No Preference (n=11)	"Difficulty" interpreted as a more clinical and defined term; n=2 referred to it as more "intrinsic". Better captures severity of cognitive issues. "Trouble" seen as too casual and vague for health assessment; n=1 referred to it as "extrinsic" (something you need to overcome), or more negatively worded than difficulty. Those who preferred "trouble" mentioned it could be easier for people to understand.	"I think difficulty is easier to understand. Trouble feels like trouble. I don't even want to answer that. I don't want to say trouble." (69F, Caregiver) "Trouble seems more appropriate. I think more people would understand." (#017,37F, Care User) "I think trouble is a simpler word. I don't think it really means anything different between the two [trouble and difficulty]." (Chronic Illness)
"Lacked control over your day-to-day life" (M) vs "No control over your day-to-day life" (E))	No=Lack (n=4) No≠Lack (n=28) Missing (n=2)	No (n=6) Lack (n=20) No Preference (n=6) Missing (n=2)	No control was seen as more extreme or absolute; n=3 participants noted that 'no control' is more suitable as a binary yes/no question. 'Lack control' allows more nuance and is more realistic (people always have some control over choices they can make) and is more consistent with the response options	

Modification of item order: Activities first (M) vs Mobility first (E)	n/a	Activities First (n=17) Mobility First (n=5) No Preference (n=11) Missing (n=1)	Participants in favor of asking about mobility first endorsed that day to day activities depend on mobility; other used the same logic to support asking about day-to-day activities first. Ultimately, no participant had very strong opinions about this.	"The order of them doesn't matter, because they're both being inside and outside of coincide with day-to-day anyway." (#005, 59 F, 60)
Modification of response option wording: "A little of the time" (M) (vs "Only occasionally" (E))	A little of the time = Only Occasionally (n=15) A little of the time ≠ Only Occasionally (n=16) Unclear or Missing (n=3)	A little of the time (n=15) Only occasionally (n=9) No Preference (n=10)	"A little of the time" seen as more quantifiable, more specific and clearer, with better distinction from "sometimes", and more consistent with the 7-day timeframe (only occasionally might work with a longer time period e.g., 6 months) "Only occasionally" was viewed as vague and broad; too similar to "Sometimes". Some preferred the brevity compared to "a little of the time", or they did not have a preference.	"I've never been a fan of occasionally. It doesn't seem enough to me. I know it's a preference. But a little seems very clearly like a week." (#004, 25 F, 60) "But occasionally oh, on occasion: everybody has that means." (#021, 69 F, 60)
Modification of response option wording: " Moderate difficulty" (vs " Some difficulty")	Moderate = Some (n=7) Moderate ≠ Some (n=19) Unclear or Missing (n=8)	Moderate (n=17) Some (n=7) Unclear or Missing (n=10)	"Moderate" provides clearer severity indication. Better aligns with clinical assessment, fits better with the other response options; and helps with response discrimination ("some" seen too similar to "slight" response level) "Some" too vague and subjective, although preferred by select individuals who suggested it may be more understandable to a majority of people.	"Moderate, is more like middle, like average." (#32NB/O, Social Care User) "Everyone may not know what moderate means, but you know some means versus a little." (#47 F, Chronic Illness)

			From those who saw them differently, moderate seemed more severe than some.	
Modification of layout: Single shared stem vs repeated question text for items	n/a	no stem (repeating the question) (n=11) Item Stem (n=5) No preference (n=7) Missing (n=11)	Those who preferred no stem found it clearer and easier to follow - they did not have to remember what the stem was asking while reading each item. Those who preferred having an item stem appreciated reduced redundancy and structure.	“I kind of like the bottom [with repeated stem] because it puts the emphasis on the item as opposed to like the top where you're thinking more about the stem. It's difficult a day to day as opposed to like what do you did.” (#002, 33 F, Chronic Illness Caregiver) “it [shared stem] reduced redundancy. So, it's easier.” (#002, 33 F, Chronic Illness Caregiver)